

Retail Spatial Intelligence

CCTV Foot-Traffic & Spatial Behavior Report

VERIFIED COMPUTER VISION DATA

Analysis Parameters & Metadata

VIDEO SOURCE:	sample_retail_cctv.mp4	DATE PROCESSED:	Oct 6, 2026, 09:24 PM
RESOLUTION:	768x432 (768x432)	VIDEO DURATION:	49.67 seconds (596 frames @ 12 fps)
ZONE CONFIG:	Automatic Camera Regions	REPORT VERSION:	v2.1 (Canonical Analytics v2.0)
TRACKING QUALITY:	Moderate Confidence (High Confidence)		
ANALYSIS ID:	1fe3a046-da71-4331		

UNIQUE VISITORS

7

Valid tracked customers

AVG ZONE DWELL

3.42s

Median: 4.5s

PEAK OCCUPANCY

3

Concurrent shoppers

TOTAL ZONE VISITS

9

Across 4 zones

Executive Summary

DETERMINISTIC SPATIAL RULES SUMMARY

Test executive summary.

1. Computer Vision Processing & Traffic Diagnostics

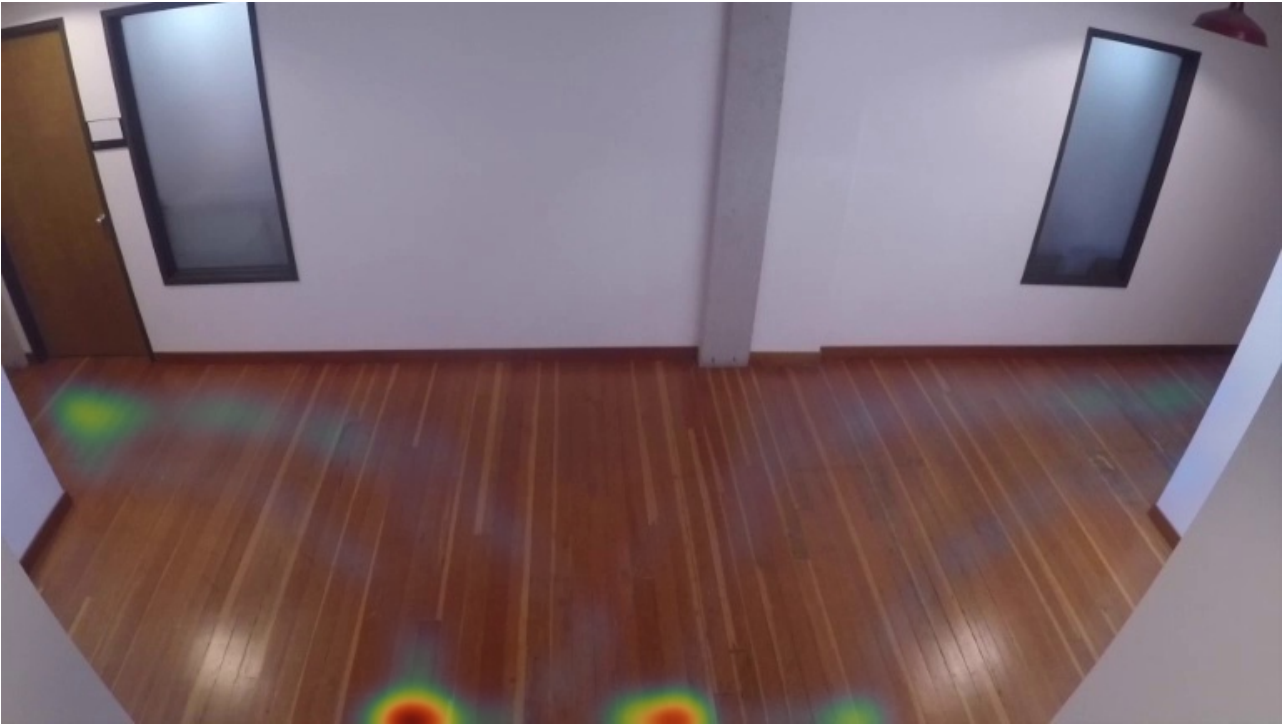
Footage was processed using Ultralytics YOLOv8 for human detection and ByteTrack for continuous anonymous multi-object tracking. Detection ground points were extracted at the bottom-center of bounding boxes ($x = [x1+x2]/2$, $y = y2$) to represent true floor contact. Tracks below the duration threshold of 1s were filtered to remove transient occlusion artifacts.

PROCESSING TELEMETRY

Raw Tracks Detected: 7	Effective Tracking Confidence: Moderate Confidence	Avg Detection Confidence: 0.83
Filtered Transient Tracks: 0	Average Track Duration: 2.3s	Relative Speed: 99 px/s

2. Observed Foot-Traffic Density Heatmap

Heatmap generated via 2D Gaussian Kernel density accumulation of verified customer foot-point coordinates overlaid onto the representative video frame.



Heatmap Methodology & Legend:

Blue = Low Pedestrian Activity | Green/Yellow = Moderate Dwell & Flow | Red/Crimson = Peak Traffic & Congestion Stagnation

3. Spatial Zone Performance Analysis

Deterministic measurements per defined spatial zone. Dwell sessions include a 0.8s occlusion buffer to prevent transient drops from breaking sessions.

Zone Name	Visits	Unique	Avg Dwell	Max Dwell	Peak Occ	Avg Occ	Share (%)	Density
Region 1 (Upper Left)	0	0	0s	0s	0	0	0%	0
Region 2 (Upper Right)	0	0	0s	0s	0	0	0%	0
Region 3 (Lower Left)	5	5	3.5s	4.83s	2	0	0%	6.03
Region 4 (Lower Right)	4	5	3.33s	4.83s	2	0	0%	6.03

4. Customer Flow & Traversal Pathways

Zone-to-zone transitions mined from consecutive spatial crossings across verified customer paths:

No multi-zone transitions observed during the sample period.

Frequent Customer Journey Sequences:

1. Region 4 (Lower Right) !' Region 3 (Lower Left)

Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 0s
2. Region 3 (Lower Left)

Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 0s
3. Region 4 (Lower Right)

Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 0s
4. Region 3 (Lower Left) !' Region 4 (Lower Right)

Observed in 1 customer(s) (14.3% of journeys) - Avg Duration: 0s

5. Forensic Bottleneck & Dead-Zone Intelligence

POTENTIAL BOTTLENECK ANALYSIS:

- **Region 3 (Lower Left) (Bottleneck Score: 80.4/100 - Confidence: High)**
 - Evidence: Peak simultaneous occupancy reached 2 visitors
 - Evidence: Elevated dwell time of 3.5s exceeds store average (3.42s)
 - Evidence: Movement speed (96.8 px/s) slowed by 2% relative to store baseline (99.0 px/s)
 - Evidence: High traffic density index (6.03 visitors per 100k sq px)
- **Region 4 (Lower Right) (Bottleneck Score: 78.5/100 - Confidence: Moderate)**
 - Evidence: Peak simultaneous occupancy reached 2 visitors
 - Evidence: High traffic density index (6.03 visitors per 100k sq px)

POTENTIAL DEAD-ZONE ANALYSIS:

- **Region 1 (Upper Left) (Deficiency Score: 100/100, Activity: 0% - Confidence: High)**
 - Evidence: Zero customer visits recorded during observed footage
 - Evidence: Only 0.0% customer penetration (0 unique visitors entered)
 - Evidence: Low peak occupancy: maximum 0 person at any moment
- **Region 2 (Upper Right) (Deficiency Score: 100/100, Activity: 0% - Confidence: High)**
 - Evidence: Zero customer visits recorded during observed footage
 - Evidence: Only 0.0% customer penetration (0 unique visitors entered)
 - Evidence: Low peak occupancy: maximum 0 person at any moment

HIGH-ACTIVITY DESTINATION ZONES:

- Region 3 (Lower Left): 5 visits, 3.5s avg dwell, peak concurrent: 0 (High Activity)
- Region 4 (Lower Right): 4 visits, 3.33s avg dwell, peak concurrent: 0 (High Activity)

6. Strategic Recommendations & Priority Action Plan

Prioritized Action Plan:

7. Scientific Methodology & Audit Traceability

- YOLOv8 Person Detection**
Pretrained neural model detecting COCO Class 0 (person). Frame sampling optimizes CPU throughput without altering physical timeline.
- ByteTrack Multi-Object Tracking**
Associates detections using high/low confidence thresholds and Kalman state estimation with persistent anonymous IDs (Customer #1, #2).
- Floor Foot-Point Extraction**
Computes ground contact coordinate at $x = (x1+x2)/2$, $y = y2$ to bypass perspective bounding box tilt.
- Point-in-Polygon Zone Assignment**
Assigns floor foot-points to arbitrary user-drawn polygons or balanced camera regions using OpenCV pointPolygonTest.
- Dwell Time with Occlusion Grace Buffer**
Maintains an 0.8s tolerance buffer before closing dwell sessions to prevent brief tracking drops from artificially splitting sessions.
- 2D Gaussian Density Heatmaps**
Accumulates foot-points into a spatial matrix smoothed with a normalized Gaussian kernel and mapped through OpenCV Turbo colormap.

8. System Limitations & Forensic Transparency

- Single-Camera Monocular Perspective: Field of view is restricted to the camera mounting angle. Speeds are reported in image-space pixels/second, not calibrated metric coordinates.
- Visual Occlusion: Dense crowds or structural pillars can temporarily occlude foot points. The platform uses a 0.8s tolerance buffer, but long occlusions may cause track fragmentation.
- Zero PII Policy: The system operates with transient anonymous IDs only. No face recognition, demographic profiling, or biometric storage is performed.
- No POS Integration: Analysis reflects physical pedestrian movement only. Purchase conversions, basket size, and revenue metrics require external POS pairing.

AUTHENTICATION & VERIFICATION STAMP

Generated deterministically by Retail Spatial Intelligence Platform Engine v2.1. All metrics directly traceable to video frame detections. Job ID: 1fe3a046-da71-4331-ae2d-cc6c980f9440

